# Practice workbook · Exponential equations, including hidden quadratics

Equations with the unknown in a power, solved by matching bases or taking logs, including those that become a quadratic or cubic in a substitution such as u = e^(3x) or u = 2^y.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-09

# 1. Question 11 · part a.ii · 3 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part a.ii only.

**11**

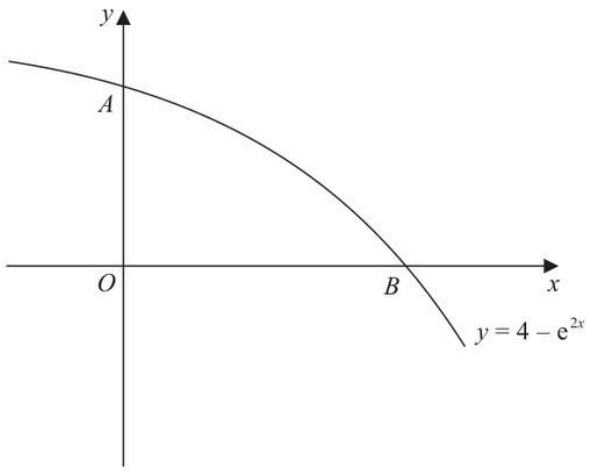



Diagram **NOT** accurately drawn

**Figure 3**

Figure 3 shows part of the curve $C$ with equation $y = 4 - e^{2x}$
The curve $C$ crosses the $y$-axis at the point $A$ and the $x$-axis at the point $B$.

(a) (i) Write down the $y$ coordinate of point $A$.

(ii) Show that the $x$ coordinate of $B$ is $x = \ln 2$

**(3)**

The line $l$ is the normal to $C$ at the point $B$.

(b) Find an equation for $l$, giving your answer in the form $y = mx + c$

**(4)**

The finite region $R$ is bounded by $C$, $l$ and the $y$-axis.

(c) Using calculus, find the area of $R$.
Give your answer to one decimal place.

**(7)**

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## 1. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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## 2. Question 11 · part a.ii · 3 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

**11**

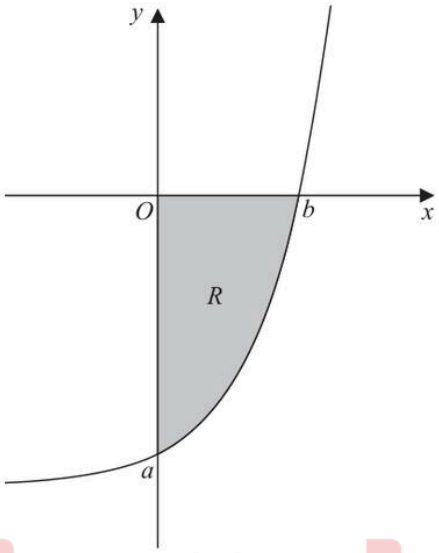



Diagram **NOT** accurately drawn

**Figure 3**

The finite region $R$, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of $a$

(ii) Show that $b = \ln 3$

**(3)**

The region $R$ is rotated through 360° about the $x$-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where $p$ and $q$ are integers.

**(6)**

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## 2. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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## 3. Question 10 · part a · 4 marks

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Further practice · Attempt part a only.

**10** (a) Show that $\frac{9^{3y}}{243} = 3^{(6y-5)}$

(4)

(b) Solve the simultaneous equations

$$\frac{9^{3y}}{243} = 27^{(x-2)}$$

$$\log_{10}\sqrt{6xy} = \log_4 2$$

(9)

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## 3. Question 10 · part a · 4 marks · Working and answer

Working and answer

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## 4. Question 8 · part a · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

**8** The curve $C_1$ has equation $y = 5e^{-2x} + 4$

The curve $C_2$ has equation $y = e^{2x}$

The curves $C_1$ and $C_2$ intersect at the point $A$.

(a) Find the exact coordinates of $A$.

**(4)**

The tangent at $A$ to $C_1$ intersects the $x$-axis at the point $B$.

(b) Show that the $x$ coordinate of $B$ is $\frac{1}{2}(5 + \ln 5)$

**(5)**

The tangent at $A$ to $C_2$ intersects the $x$-axis at the point $D$.

(c) Find the area of $\Delta ABD$.

**(6)**

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**20**

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## 4. Question 8 · part a · 4 marks · Working and answer

Working and answer

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## 5. Question 8 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

**8**

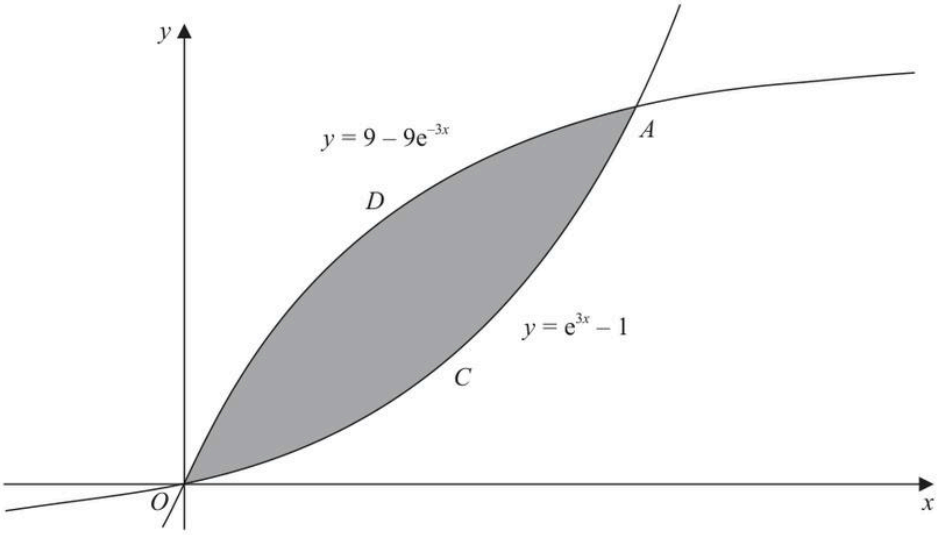



**Figure 2**

Figure 2 shows part of the curve $C$ with equation $y = e^{3x} - 1$ and part of the curve $D$ with equation $y = 9 - 9e^{-3x}$

The curves intersect at the origin $O$ and the point $A$.

(a) (i) Show that the $x$ coordinate of the point $A$ satisfies the equation

$$(e^{3x})^2 - 10e^{3x} + 9 = 0$$

(ii) Hence, show that the $x$ coordinate of the point $A$ is $\frac{1}{3}\ln 9$

**(5)**

The finite region bounded by $C$ and by $D$ is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

**(6)**

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## 5. Question 8 · part a.i · 0 marks · Working and answer

Working and answer

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## 6. Question 8 · part a.ii · 5 marks

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Further practice · Attempt part a.ii only.

**8**

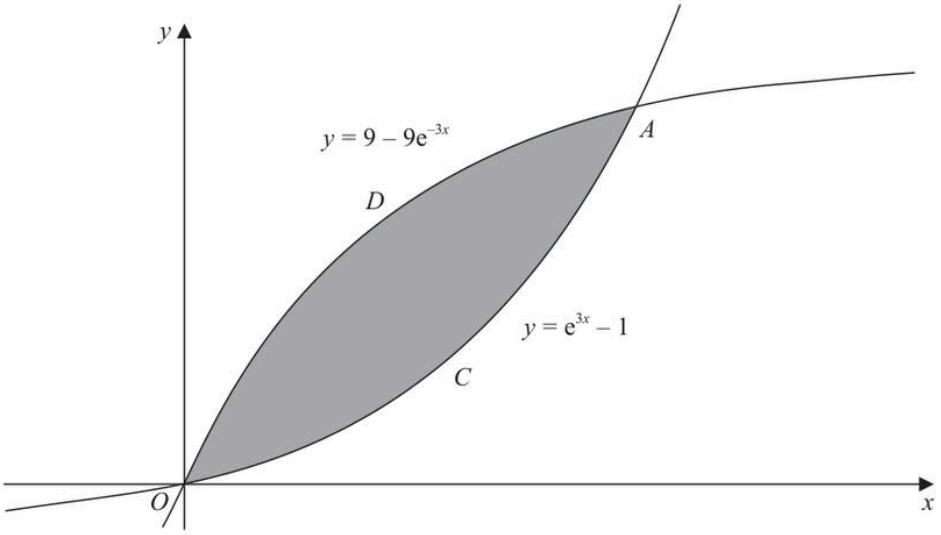



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The finite region bounded by $C$ and by $D$ is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

**(6)**

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P 7 1 6 4 2 A 0 2 2 3 6

## 6. Question 8 · part a.ii · 5 marks · Working and answer

Working and answer

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## 7. Question 5 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

**5** $f(x) = 2ax^3 + x^2 - bx + 3a$ where $a$ and $b$ are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of $b$

**(5)**

(b) Hence factorise $f(x)$ completely.

**(2)**

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

**(5)**

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## 7. Question 5 · part c · 5 marks

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Further practice · Attempt part c only.



## 7. Question 5 · part c · 5 marks · Working and answer

Working and answer

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